

Talk Like a Structured Graph? Structural Statistics Help Small LLMs Mainly by Stating the Answer

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Abstract

As language models are increasingly used to reason over structured data, we ask whether explicit structural features, which benefit graph neural networks, also help a frozen language model execute queries over a textual graph. We prepend node degree, local clustering coefficients, and random-walk return probabilities to GraphQA’s incident encoding and evaluate Qwen3-1.7B and Qwen3-4B, with and without thinking, across 84,000 responses on 700 paired graphs. Stated degrees improve node-degree accuracy for both Qwen3-1.7B modes but reduce it for plain Qwen3-4B, which already counts accurately. This does not establish better graph reading: a graph-blind solver that never reads the edge list answers degree and edge-count queries at 100% accuracy, against its own 14.4% and 3.0% without a primer. Nor are stated answers always used: with its degree printed, plain Qwen3-1.7B is correct on 43% of degree prompts, against 39% without. Clustering and return probabilities, which do not state the answer, have inconsistent effects and no significant gain in joint degree and neighbor-set accuracy. Correct answers need not rest on the graph: 43.5% of plain Qwen3-1.7B’s correct cycle answers without a primer cite a nonexistent edge. Explicit statistics can help by supplying answers or changing how a model uses the prompt, but do not consistently improve graph-query execution.¹

1 Introduction

Structural features and positional encodings improve graph neural networks and graph Transformers by supplying information that message passing may not readily recover (Dwivedi et al., 2022; Rampásek et al., 2022). A language model given a graph as text must instead extract structure from node and edge descriptions. In *Talk like a Graph*, Fatemi et al. (2024) report that LLMs find basic graph tasks difficult and that performance depends

on the encoding, task, and graph structure; NL-Graph likewise finds that prompting becomes less effective on more complex problems (Wang et al., 2023). We ask: *when a black-box language model reads a textual graph, do explicit structural statistics improve execution of graph queries?*

Prior work varies how graphs are serialized or supplies learned graph tokens and structural prompts (Section 2). These studies leave a narrower question open: whether precomputed statistics help a model answer graph queries when the textual edge list and question are fixed. Answering it requires care: a degree sentence can itself contain the answer to a degree query, so an accuracy gain need not indicate better use of the edges.

We test this question by placing a primer of degree, local clustering coefficients, or random-walk return probabilities before GraphQA’s incident encoding (Figure 1). We evaluate Qwen3-1.7B and Qwen3-4B, with and without thinking, on paired graph-question prompts across six GraphQA tasks. We also test the combined statistics, a component-count sentence, and a structure-free filler. A graph-blind solver, which never reads the edge list, gives a lower bound on what the primer makes available; paired comparisons, joint degree and neighbor-set correctness, and checks of response claims against the graph provide further evidence about how answers are obtained.

Our contribution is a controlled evaluation of these statistics as textual primers for graph queries, with controls for answer supply and the effects of added text. This framework examines whether a change in accuracy reflects information given directly by a primer or improved use of the graph’s incident encoding. We find that stated degrees help mainly by supplying the answer, while statistics that do not supply it change answers inconsistently.

¹<https://github.com/GalBrk/GraphTalk>

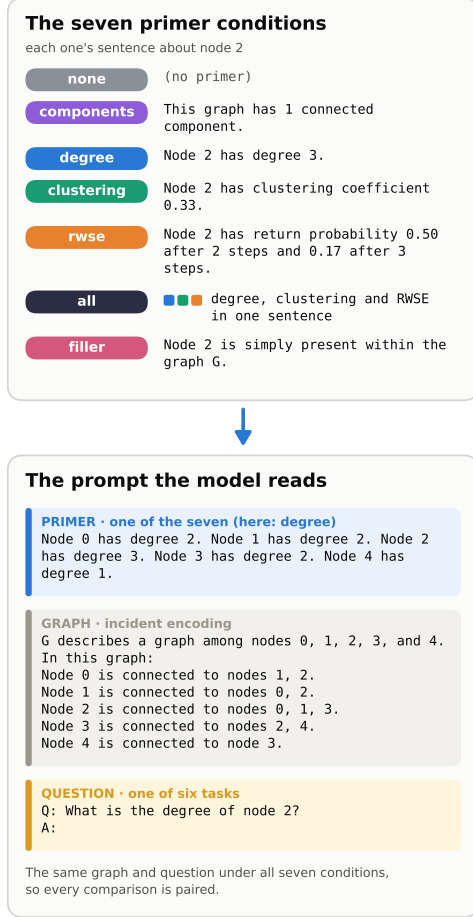


Figure 1: How a primer is used. Top: the seven conditions, each with its sentence about node 2 of a five-node example (the study’s graphs have 40 nodes). Bottom: the prompt the model reads, one primer followed by GraphQA’s incident encoding and the question. The degree and all primers state the answer to this question.

2 Related Work

Graph representations for LLMs. *Talk like a Graph* introduces GraphQA and studies how textual encodings, prompts, and graph structure affect graph-query accuracy (Fatemi et al., 2024); NL-Graph evaluates natural-language graph problems and prompting methods (Wang et al., 2023). Other approaches supply learned graph tokens (Perozzi et al., 2024), motif descriptions (Das et al., 2024), or structural context for node classification on text-attributed graphs (Huang et al., 2024). Our experiment holds the incident encoding and question fixed while varying named statistics in plain text, including whether a statistic supplies the queried answer.

Controls and response evidence. Added context, input length, and formatting can change model

answers (Shi et al., 2023; Levy et al., 2024; Sclar et al., 2024). A successful partial-input baseline exposes a shortcut, but an unsuccessful one cannot exclude other shortcuts (Feng et al., 2019). Generated explanations may also misrepresent how an answer was reached (Turpin et al., 2023). We therefore compare primers with a structure-free filler, measure what a graph-blind solver can recover, and check stated graph claims against the actual edges.

3 Method

Models and graphs. We evaluate Qwen3-1.7B and Qwen3-4B (Qwen Team, 2025), each with thinking off (P) and on (T); each model–mode pair is an *arm*. We use zero-shot greedy decoding and one response per prompt (budgets in Appendix A). A pilot on published GraphQA graphs left larger models near ceiling (Appendix C). We therefore use 40-node independent-edge graphs $G(n, p)$ (Gilbert, 1959) at $p \in \{.10, .20, .35, .50\}$, extending node-degree and edge-existence tests to $p \in \{.65, .75, .85\}$. Every condition and arm receives the same 100 graphs per task and density, allowing paired comparisons.

Tasks and primers. We retain GraphQA’s incident encoding and six tasks (Figure 1). Answers to node degree, connected nodes, edge count, and edge existence vary across graphs. The other two tasks have constant answers: every graph has 40 nodes and contains a cycle, so we use cycle-check responses to assess the support given for a “yes” answer. We score connected nodes primarily by exact match of the predicted and true neighbor sets; set-F1 is a secondary measure that gives partial credit.

Before the encoding, we place either no primer; one sentence per node stating its degree, local clustering coefficient, or two- and three-step random-walk return probabilities (a truncated random-walk structural encoding, RWSE; Dwivedi et al., 2022; Rampásek et al., 2022); or a sentence combining all three statistics. We add two controls: a sentence giving the number of connected components, and a filler that names each node without stating a statistic. The filler is similar in length to clustering but not to the other primers (Appendix Table 2). A rule-based graph-blind solver tests what can be answered from the primer without the edge list; it also receives the node and edge counts but never returns them as answers.

Task	Arm	none	degree	clustering	RWSE	all	components	filler
Degree	1.7P	60.50	+7.5 ^f	+3.0	+4.0	+10.2 ^f	+1.8	-0.5
	1.7T	76.25	+8.8 ^f	+1.5 ^f	+2.5 ^f	+12.2 ^f	-1.0	-3.0
	4P	99.25	-6.5 ^f	-0.5	-2.2	-8.0 ^f	+0.2	-0.5
	4T	97.00	+2.5	+1.8	-1.2 ^f	+0.5	+0.2	+2.0
Neighbors	1.7P	72.50	+3.5 ^f	+1.2 ^f	+1.0 ^f	+2.0 ^f	+0.8 ^f	-6.2
	1.7T	78.75	+6.2	+2.0	+0.2	+6.2	+2.5	+2.5
	4P	86.00	-3.0 ^f	+1.0 ^f	+0.0 ^f	-2.8 ^f	+9.2 ^f	-12.5
	4T	95.50	+0.5	-0.8	+0.5	-1.2	+0.5	+1.2
Edge count	1.7P	1.25	+3.0 [†]	-0.5 [†]	-0.8 [†]	+0.8 [†]	-1.0 [†]	-0.5 [†]
	1.7T	15.75	-3.2 [†]	-1.2 [†]	-3.2 [†]	+5.0 [†]	-0.8 [†]	-1.5 [†]
	4P	2.00	+27.2 ^f	-1.2 ^f	-1.5 ^f	+3.8	-1.2 ^f	+1.2
	4T	20.00	+8.5 [†]	+24.8 [†]	+10.2 [†]	+1.8 [†]	+24.8 [†]	+12.2 [†]
Edge exists	1.7P	70.50	+4.0 ^f	+3.8 ^f	+5.0 ^f	+11.5 ^f	-8.2 ^f	-15.8
	1.7T	99.00	-2.5	-1.0	-1.2	-1.8	-0.8	-0.5
	4P	90.50	-4.8	+4.0 ^f	-10.0	-10.0	-2.8 ^f	-7.0
	4T	99.75	+0.0	-0.8	-1.2	-1.0	-0.5	+0.0

Table 1: Primary result: correct share without a primer (% of all prompts) and paired change under each primer (points), on the same 400 graphs per row ($p \leq .50$). Shaded: the primer supplies the degree answer or every term needed to compute edge count. Bold: $q < .05$ against no primer (BH within arm, task and control); ^f: $q < .05$ against filler; [†]: at least 15% truncation on either side, and no significance marks. Intervals for selected contrasts: Table 3; truncation: Table 5; balanced accuracy: Table 6. P/T: plain/thinking.

Outcomes and inference. Each response is classified as correct, wrong, or truncated; accuracy uses all prompts as its denominator. The primary effect is the paired change in accuracy, in percentage points, against no primer and against filler on the 400 main-sweep graphs per task (Table 1). We use exact McNemar tests (McNemar, 1947), bootstrap intervals over graphs, and Benjamini–Hochberg correction (Benjamini and Hochberg, 1995) within each arm–task–control family. Cells with at least 15% truncation in either condition are flagged and excluded from pooled summaries. We also report balanced accuracy for edge existence because its positive-answer rate varies with density. Plain–thinking comparisons use the same graphs; we call the modes equivalent only when the 90% interval for their difference lies within ± 3 points. A non-significant effect is not evidence of no effect (detectable sizes in Appendix A).

Evidence of graph use. Because degree and neighbor-set questions target the same node, their joint correctness tests whether a degree gain accompanies correct identification of that node’s neighbors; it does not establish the model’s internal procedure. Rule-based classifiers (Appendix A) identify direct-answer and neighbor-list response patterns, primer conflicts, and graph claims. For correct, finished cycle-check answers, we distinguish a cited cycle whose steps are all edges, a claimed cycle containing a nonexistent edge, a walk that is not a cycle, and an answer without an explicit cycle. Within the last category, we also identify valid

arguments from edge count: a 40-node graph with at least 40 edges must contain a cycle. Conflict detection was compared with blind hand labels; the other classifiers were spot-checked by an LLM under human supervision.

Follow-ups. We repeat Qwen3-1.7B node-degree tests on 400 graphs per density across all seven densities and run fresh-seed and 20–160-node checks at mean degrees 8 and 16; the size grid also includes Qwen3-8B.

4 Results

4.1 What the primers make available

From stated degrees alone, the graph-blind solver recovers node degree directly and edge count by halving the degree sum, both with 100% accuracy. Without degrees, the rule-based solver reaches at most 21.0% on degree and 10.4% on edge count, compared with 14.4% and 3.0% without a primer (Appendix Table 2). On sparse graphs, however, a lookup fitted on separate graphs recovers degree from rounded clustering or RWSE values substantially better than the majority answer (Appendix A). In the models’ responses, the largest main-sweep gains on the four variable-answer tasks occur when degrees supply the answer or its terms; primers without degrees and the controls produce effects of either sign (Table 1; Appendix Figure 2).

4.2 Stated degrees can help or disrupt counting

The degree primer raises node-degree accuracy by 7.5 points for plain 1.7B and 8.8 for thinking 1.7B, but lowers it by 6.5 for plain 4B, whose unaided accuracy is 99.25% (Table 1). Its effect depends on the cost of counting from the incident list. Within plain 4B, it harms queries about the lowest-degree nodes and helps those about the highest-degree nodes (-11.8 and $+10.6$ points across seven densities); at $p = .85$, where unaided accuracy falls to 33.0%, it gains 27.0 points (Appendix B). Stated degrees also supply every term needed for edge count, raising plain 4B’s main-sweep accuracy from 2.00% to 29.25%, far below the solver’s 100%.

Stated answers are not always used. Across seven densities, plain 1.7B is correct on 43% of degree prompts under the degree primer, against 39% without it; none of the answers the primer turns from wrong to correct at $p = .20$ and .35 cites the stated value. Plain 4B, by contrast, appears to look the answer up: at $p = .50$, 62% of its responses answer directly and only 8% list the neighbors; across densities, its wrong lookups match the degree printed one sentence away more often than chance (Appendix B). Its losses under the degree primer are concentrated on prompts where the response pattern changes.

4.3 Primers also change answers they do not supply

Filler alone flips up to 18.6% of an arm’s outcomes against no primer, showing that added text can matter without adding a statistic (Appendix B). Clustering and RWSE mostly have smaller main-sweep effects than degree, varying in sign by task and model. On plain 4B’s edge-existence queries, clustering raises balanced accuracy by 3.0 points while RWSE lowers it by 6.5 (Appendix Table 6). Clustering’s 3.0-point gain on plain 1.7B degree queries is not significant in Table 1 but replicates on unscreened graphs and a fresh seed; it is not detected at $p \geq .65$ or in Qwen3-8B (Appendix C). Combining the statistics can reverse their separate effects: on plain 4B’s dense degree queries, degree, clustering, and RWSE each raise accuracy (9.3, 11.3, and 7.7 points), but the combined primer lowers it by 21.3 on the same graphs (Appendix B).

On the four variable-answer tasks, the largest unflagged main-sweep gain without degrees is the

component sentence’s 9.2-point increase in plain 4B’s *exact* neighbor-set accuracy; its set-F1 barely moves (Appendix B). The sentence is identical across all sampled graphs at $p \geq .20$, yet responses restate the queried node’s incident line much more often with it than with no primer or filler (98.5%, 63.0%, and 21.8%). Filler moves this response pattern in the opposite direction and lowers exact-match accuracy (Table 1).

4.4 Can a primer close the gap to thinking?

Where thinking outperforms plain mode, primers rarely close the gap, and for plain 4B several widen it (Appendix Table 8). Among the exceptions, the component sentence brings plain 4B’s neighbor-set exact match within the ± 3 -point equivalence margin of thinking 4B without a primer, while filler widens that gap; the combined primer reduces plain 1.7B’s node-degree gap from 15.8 to 5.5 points; and under degree, plain 1.7B’s neighbor-set gap is not significant, though its interval extends beyond the margin.

The degree primer puts plain 4B ahead of thinking 4B on edge count, but the comparison mostly reflects completion: thinking 4B is correct on 97.6% of the edge-count responses it finishes, while 79.50% of its responses exhaust the token budget.

4.5 Primers in thinking mode

Thinking 4B is near ceiling on degree, neighbor sets, and edge existence; every primer changes these accuracies by at most 2.5 points, which says little about harder cases. Thinking 1.7B gains from stated degrees but often lists neighbors and checks its count against the printed value. This checking raises its node-degree truncation rate from 1.1% to 7.6% across seven densities; when its count and the printed value conflict, it ends with the stated value in 66% of finished cases. Truncation also shapes thinking 4B’s results: in every comparison, most changed answers involve a response that ran out of tokens (Appendix B), so some apparent primer effects reflect whether a response finishes.

4.6 Do correct answers have graph support?

Clustering and RWSE do not significantly improve the joint correctness of degree and neighbor-set answers for the same node in any arm (Appendix B). The degree primer raises this measure by 8.5 points for plain 1.7B but lowers it by 9.2 for plain 4B; the component sentence raises plain 4B’s by 9.2.

Because cycle-check accuracy cannot measure cycle detection here (Section 3), we instead examine the support in correct, finished “yes” responses (Appendix Table 10). Without a primer, 43.5% of plain 1.7B’s correct, finished responses cite an invented cycle containing a nonexistent edge. Counting over all prompts, valid cited cycles or edge-count arguments support 34.2% and 58.5% for plain 1.7B and plain 4B, respectively. The degree primer shifts plain-model responses toward valid edge-count arguments, which need no named cycle; plain 1.7B’s share of prompts with valid support rises to 50.2%, although its cycle-check accuracy falls by 17.5 points as truncation rises by 8.2. Thus, a correct “yes” can be accompanied by a valid proof, an invalid claimed cycle, or no explicit cycle; the label alone does not distinguish them.

5 Discussion

Structural statistics serve a different role here than in graph models, where random-walk encodings can distinguish structures that some message-passing networks cannot represent (Xu et al., 2019; Dwivedi et al., 2022). In our setting, the incident encoding already contains every edge, but the language model must find the relevant edges and reason over them. Stated degree can bypass that computation for degree and edge-count queries, but it also introduces a lookup that can fail when the model selects the wrong node’s sentence. Whether it helps therefore depends partly on the model’s unaided accuracy and its ability to use the stated value.

Clustering and RWSE, by contrast, do not directly answer the studied queries, and their mixed effects, together with changes caused by filler and the near-constant component sentence, are consistent with primers changing how models use the prompt. The independent-edge graphs provide limited opportunity for these statistics to convey task-relevant structure beyond degree and density; the results do not establish how they would behave on graph families where triangles or return patterns are more informative.

Finally, answer accuracy and the stated justification measure different things. Joint degree and neighbor-set correctness gives evidence about use of the incident line, while cycle checks reveal whether a correct “yes” is supported by a valid graph argument.

6 Future Work

Corrupting or omitting the queried node’s stated degree would separate following a stated value, checking it and counting from edges. Shuffled node labels and primer order, and neutral text matched in length and position, would separate presentation from information for the component sentence and clustering’s degree gain on plain 1.7B. Acyclic graphs, and families that vary triangles and return patterns at matched size and density, would give cycle check two answers and let clustering and RWSE carry information. Other model families, larger budgets and a graph-counting tool would test generality and when a primer is the cheapest way to supply a computation.

7 Conclusion

In paired GraphQA tests, stated degrees produced the clearest gains, chiefly where the model counted unreliably but could use the stated value; they could also disrupt accurate counting. Clustering and random-walk returns produced task- and density-dependent effects without a consistent gain in joint degree and neighbor-set correctness. Controls and graph-checked responses showed why accuracy alone is insufficient: added text can change answers, and even a correct cycle label can cite a nonexistent edge. The value of a textual statistic depends on the query, what the statistic gives away, and what the model can already compute.

Limitations

We test two Qwen3 models (Qwen3-8B in one follow-up), one incident encoding, fixed prompts, and mostly 40-node independent-edge graphs. All sampled graphs contain cycles, so cycle analysis evaluates support for “yes” answers, not cycle discrimination. The filler only roughly matches clustering’s length, and component count supplies structural information. The graph-blind solver cannot exclude other clues. We did not test an instruction-only control; the queried node’s list position and identifier length are confounded with each other. Fixed budgets cause substantial truncation in some thinking-mode cells, and reruns can differ. Response categories describe behavior, not internal computation: only conflict classifiers were checked against blind hand labels; the others received supervised spot checks.

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A Method Details

Primer statistics. For the undirected adjacency matrix A , let $d_v = \sum_u A_{vu}$ and $P_{vu} = A_{vu}/d_v$ when $d_v > 0$ (a zero row otherwise). If t_v is the number of edges among v ’s neighbors, local clustering is $c_v = 2t_v/[d_v(d_v - 1)]$, set to zero for $d_v < 2$. The two return features are $r_v^{(k)} = (P^k)_{vv}$ for $k \in \{2, 3\}$. Floats are printed to two decimals, with one sentence per node in node order (Table 2).

Feature choice. Degree and random-walk encodings are used in graph Transformers (Rampášek et al., 2022); clustering records edges among a node’s neighbors. We use return steps 2 and 3: step 1 is zero in a simple graph, while step 3 reflects triangles. We omit Laplacian coordinates because their signs and degenerate eigenspace bases require a convention (Dwivedi et al., 2022). Component count is a global control, though it is informative in principle: a graph with n nodes, m edges and c components is acyclic exactly when $m = n - c$.

Models, budgets and inference. We run the Qwen3 checkpoints with transformers. The output limit is 8,192 tokens, except 2,048 for plain arms at $p \geq .65$; the shorter limit costs little, because, pooled over primers, no arm and task in the high-density extension truncates more than 2.5%. Because Table 1 compares 400 paired graphs, its non-significant cells rule out only moderate gains: the median detectable gain is 5.7 points in the 1.7B arms and 3.2 in the 4B arms, and 18 of thinking 4B’s 28 null comparisons sit too close to ceiling for any gain to be detectable. A single 100-graph density cell needs 7.4 points, and side statistics (primers that do not state the queried answer) need 13.7 at 25–75% baseline accuracy. The significant results survive stricter correction: under one BH correction over all 264 tests, every bold cell of Table 1 but one, and all but three ^f marks, stay significant. Finally, even greedy decoding is not fully repeatable: rerunning 1,200 identical plain-1.7B node-degree prompts changes correctness on 5.3% of them. We therefore compare net correct shares, and report the number of responses that switch outcome only descriptively.

Graph-blind solver. Besides the printed primer and the query targets, the solver sees the graph’s node and edge counts (§3), never the incident edge list. For edge count, it computes the answer from stated degrees, estimates it from mean clustering

with a fitted rule, or otherwise falls back to the majority answer of the fitting graphs. Its 16 exact rules, one heuristic and eight fitted rules are fit and scored on disjoint graphs. Fitted rules can fail outside their range: one RWSE rule infers density from return values to answer edge-existence questions, but scores below the majority answer at high density, which is why RWSE falls below no primer on that task in Table 2. The solver’s score therefore shows what one shortcut achieves, not an upper bound on what the full prompt conveys. Sparse graphs leave strong indirect clues (§4.1): from rounded clustering values alone, a separately fitted lookup answers 51.0% and 54.0% of degree queries at $p = .10$ and $p = .20$, against majority answers of 17.0% and 16.0%.

Primer	Mean chars	Degree	Neighbors	Edge count	Edge exists
none	0	14.4	0.0	3.0	73.5
degree	891	100.0	0.1	100.0	74.6
clustering	1,629	14.4	0.0	10.4	72.7
RWSE	2,949	21.0	0.1	3.0	55.4
all	4,691	100.0	0.8	100.0	74.1
components	37	14.4	0.0	3.0	73.5
filler	1,829	14.4	0.0	3.0	73.5

Table 2: Graph-blind solver accuracy (%) on the four variable-answer tasks, averaged over seven densities ($p = .10-.85$, 40 nodes) and scored on separately generated test graphs, with mean primer length in characters (none adds no text). Components and filler score the same as no primer. The solver sees graph size and the query targets but no incident edges.

Response classifiers. A *lookup* answer does not restate the queried node’s incident line; among answers that do, an *enumeration* lists at least five neighbors on separate lines. The cycle and edge classifiers check each stated graph step against the incident encoding.

Scoring. The answer extractor parses 99.84% of responses. Neighbor-set exact match is the primary measure and set-F1 a secondary one: among the five contrasts whose exact match changes by at least 5 points, F1 moves the same way in three, at 0.11–0.17 times the size, and reverses in two. For edge existence, balanced accuracy is tested with paired permutation tests, BH-adjusted over the six conditions within each arm.

B Detailed Results

Table 3 gives intervals for selected contrasts; Table 5 reports every truncation contrast. Figure 2 groups the effects; Figures 3, 4 and 5 show degree effects across arms and densities; Figure 6 separates completion from accuracy on edge count. Other tables examine edge existence, cycle support and the plain-thinking gap.

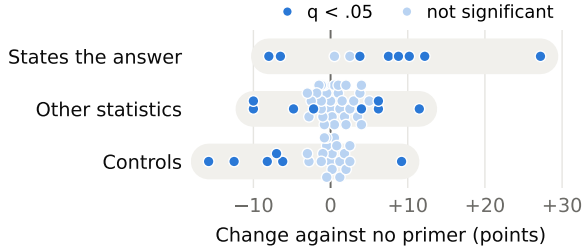


Figure 2: Unflagged main-sweep effects from Table 1, grouped by whether the primer supplies the answer, gives another statistic, or is a control. Each dot is one arm-task-primer cell; filled dots have BH $q < .05$ against no primer. The answer-stating group contains degree and all on node-degree and edge-count questions.

Task	Arm	Primer	No primer	Δ	95% CI
Degree	1.7P	degree	60.50	+7.5	[+3.0, +11.8]
	1.7T	degree	76.25	+8.8	[+4.2, +13.2]
	4P	degree	99.25	-6.5	[-9.2, -4.0]
	4P	all	99.25	-8.0	[-11.0, -5.2]
Neighbors	1.7T	degree	78.75	+6.2	[+2.8, +10.0]
	4P	components	86.00	+9.2	[+6.0, +12.8]
	4P	filler	86.00	-12.5	[-17.2, -7.8]
Edge count	4P	degree	2.00	+27.2	[+22.8, +32.0]
Edge exists	1.7P	all	70.50	+11.5	[+6.2, +16.5]
	1.7P	filler	70.50	-15.8	[-20.0, -11.8]
	4P	clustering	90.50	+4.0	[+1.0, +7.0]
	4P	RWSE	90.50	-10.0	[-13.8, -6.2]
Node count	1.7P	clustering	31.50	+67.2	[+64.5, +70.0]
Cycle	1.7P	degree	90.75	-17.5 [†]	[-23.0, -12.2]
	1.7T	degree	97.00	-21.8 [†]	[-26.5, -17.5]

Table 3: Uncertainty for selected contrasts: no-primer accuracy (% of all prompts) and paired change (points) with its graph-bootstrap interval. Every unflagged contrast shown has BH $q < .05$; [†] marks at least 15% truncation on either side. Main-sweep point estimates for all four variable-answer tasks are in Table 1; all intervals are in the repository output `outputs/n40-sweep/primer_findings.txt`. Node count and cycle check have constant gold answers. P/T: plain/thinking.

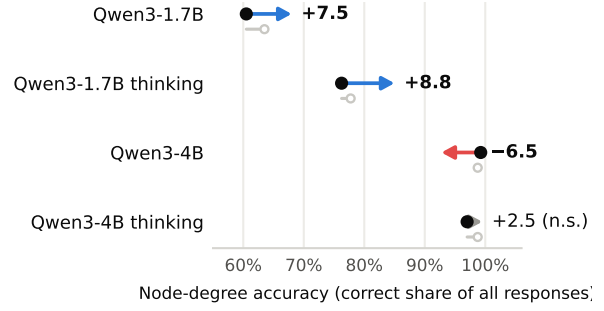


Figure 3: Node-degree accuracy per arm with no primer (dot) and with the degree primer (arrow), as in Table 1. Hollow: the clustering primer, which states no degree. Blue gains and red losses have $q < .05$ against no primer; gray does not.

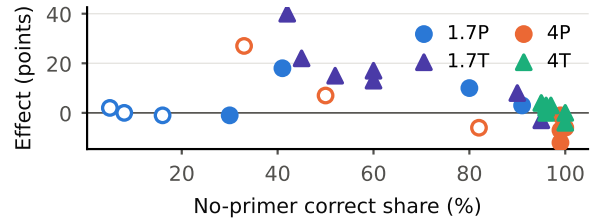


Figure 4: The degree primer's effect on node degree against the no-primer correct share, per arm and density. Hollow: plain arm at $p \geq .65$ (smaller budget).

Unaided accuracy	Answer stated	Side statistic
0-25%	+8.3 (16)	-1.7 (23)
25-50%	+16.1 (10)	+2.6 (20)
50-75%	+12.9 (8)	+3.9 (47)
75-90%	-5.8 (4)	+0.7 (51)
90-100%	-0.9 (28)	-0.9 (177)

Table 4: Mean accuracy change against no primer (points; number of cells in parentheses) by unaided-accuracy band, over unflagged arm-task-density cells of the four variable-answer tasks ($p \leq .50$, plus $p \geq .65$ for node degree and edge existence). *Answer stated*: degree and all on node degree and edge count. *Side statistic*: every other primer-task pair, including components and degree or all on neighbor sets and edge existence; filler is excluded.

Stated answers (§4.2). Answer-stating primers gain most, in points, where unaided accuracy is intermediate (Table 4; Figures 3 and 4). This holds when bands are formed on one half of the graphs and evaluated on the other, and at truncation thresholds from 5% to 50% (at least +11.6). On a log-odds scale, however, the gain is similar across the three lower bands, so the intermediate peak partly reflects the points scale; and the losses at high unaided accuracy come from plain 4B alone. Density, by contrast, has no detectable effect at a fixed unaided accuracy (in the main sweep, per-

Task	Arm	none	degree	clustering	RWSE	all	components	filler
Degree	1.7P	0.00	+0.0	+0.0	+0.2	+0.0	+0.0	+0.0
	1.7T	0.75	+4.5	-0.5	+0.0	-0.5	-0.2	-0.5
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	2.75	-2.5	-2.0	-0.5	-1.0	-0.5	-1.8
Neighbors	1.7P	0.00	+0.0	+0.0	+0.0	+0.2	+0.0	+0.0
	1.7T	1.25	+0.2	-1.0	-0.8	-0.5	-0.2	-0.8
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	3.25	-0.2	+1.2	-0.2	+1.0	+0.2	-0.5
Edge count	1.7P	29.75	-20.2	-11.8	-11.8	-1.8	-2.5	-3.5
	1.7T	81.75	-0.8	+0.8	+5.0	-12.2	+0.5	+1.2
	4P	0.00	+1.8	+0.2	+0.5	+0.8	+0.8	+0.0
	4T	79.50	-20.8	-26.0	-12.2	-23.8	-25.0	-13.0
Edge exists	1.7P	0.00	+0.0	+0.0	+0.2	+0.0	+0.0	+0.0
	1.7T	0.25	+0.2	+0.0	+0.0	-0.2	+0.0	-0.2
	4P	0.00	+0.5	+0.0	+0.0	+0.2	+0.0	+0.0
	4T	0.25	+0.0	+0.8	+1.2	+1.0	+0.5	+0.0
Node count	1.7P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	1.7T	7.75	-5.0	-7.8	-7.8	-7.8	-0.5	-7.8
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	0.00	+0.2	+0.2	+0.2	+0.0	+0.0	+0.0
Cycle	1.7P	9.00	+8.2	-8.2	-7.2	-8.0	-2.5	-2.8
	1.7T	3.00	+21.2	+0.5	-2.2	+0.8	+7.5	-3.0
	4P	0.75	+0.8	-0.2	+0.5	-0.2	+0.8	+0.5
	4T	0.75	+4.8	+0.0	-0.2	+0.5	+3.2	-0.5

Table 5: Complete main-sweep truncated shares for all six tasks: no-primer baseline (% of all prompts) and paired change (points). Together with Table 1, these give the *wrong* share: $wrong = 100 - correct - truncated$. P/T: plain/thinking; components and filler are controls. Truncation contrasts carry no significance marks.

mutation $p = .28$ for answer-stating primers and $p = .85$ for side statistics). The one exception, a side-statistic effect on node degree that falls with density ($p = .006$), disappears once $p \geq .65$ is added ($p = .58$). Plain 4B spans both ends of Table 4: besides its gain at $p = .85$, the degree primer costs it 12.0 points at $p = .50$, where it answers 99.0% unaided (both $q < .01$; Figure 4). A printed answer must still be found and used. Across seven densities, even though the solver reads every degree correctly, accuracy under the degree primer ranges from 43% for plain 1.7B to 98% for thinking 4B. At high density, the degree primer does not detectably help plain 1.7B (+1.2 points, $q = .44$); instead, at $p = .85$ the model answers 39, the degree of a node joined to all others, on 70.8% of finished responses with the primer and 61.0% without. One source of error is a neighboring line: across seven densities, 17 of 57 wrong plain-4B lookups copy the degree of an adjacent sentence, more than chance predicts ($p = .017$). The combined primer is costlier still for dense plain 4B (§4.3; $q < .001$, against $q = .03$ for degree alone’s gain); Figure 5 shows this change for every arm and density.

Controls and run-to-run noise (§4.3). Filler changes 6.2–18.6% of outcomes per arm, and the primers largely change the same questions. In every arm, every primer, components included, changes 38.5–69.1% of the questions filler changes but only 3.1–15.2% of the rest, and this gap holds within

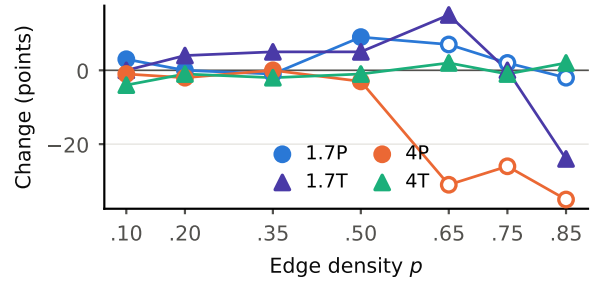


Figure 5: What adding clustering and RWSE to the same degrees changes on node degree (all three statistics minus the degree primer, points), per arm and density. Hollow as in Figure 4.

task and density ($q < .05$ for all twenty contrasts). Part of this overlap is noise. When plain 1.7B’s node-degree prompts are generated a second time, 24 of 400 questions change outcome, and every condition changes these questions more often than the rest ($q \leq .018$). However, they account for only 11–16 of each condition’s 65–113 changes, and because each prompt was generated only twice, this undercounts how many changes noise alone causes. Filler’s changes also tend to be losses: where unaided accuracy is 25–75%, components, clustering and RWSE average +1.6 points, while filler averages -6.9 on the same arm-task-density cells. On plain 4B’s neighbor sets, restatements of the queried incident line are about equally accurate with the component sentence and without a primer (95% and 92%; 84% with filler), so the sentence’s gain comes mainly from restating more often.

Edge existence (§4.3). Primers change edge existence mainly in the plain arms (Table 6). There, clustering and RWSE move plain 4B in opposite directions, and the combined primer helps plain 1.7B but harms plain 4B. Because the share of true edges changes with density, the table adds balanced accuracy, which weights a correct yes and a correct no equally; “yes share” is the fraction of finished responses answering yes.

Primer	1.7P			4P		
	Raw	Balanced	Yes	Raw	Balanced	Yes
none	70.50	79.86	56.2	90.50	93.22	35.7
degree	74.50	82.30	51.7	85.75	90.27	40.7
clustering	74.25	82.42	52.5	94.50	96.25	32.3
RWSE	75.50	82.68	50.1	80.50	86.69	46.3
all	82.00	87.71	44.8	80.50	86.69	46.1
components	62.25	74.23	64.5	87.75	91.64	39.0
filler	54.75	69.11	72.0	83.50	88.74	43.2

Table 6: Edge existence in plain arms (400 prompts per row): raw and balanced accuracy over all prompts and “yes” share among finished answers (%). Bold: balanced accuracy differs from no primer (paired permutation, BH $q < .05$ within arm). Thinking-arm balanced accuracy remains 97.61–99.83%; their full rows are in the repository output `outputs/n40-sweep/primer_findings.txt`.

False alarms rise with the number of shared neighbors (Table 7). In $G(n, p)$ that number is independent of whether the pair is joined, so this pattern suggests that models mistake proximity for adjacency. It is not a density artifact: within each density and without a primer, false alarms share on average 4.48 more neighbors than correct no’s for plain 1.7B ($p = .0005$) and 2.81 more for plain 4B ($p = .014$). Clustering changes stated reasons more than decisions: under clustering, plain 1.7B’s responses explicitly claiming a non-edge is an edge fall from 13.7% to 4.4%, but its false alarms fall only from 40.3% to 35.2%.

Plain arm	0	1	2–3	4+
1.7B	1.7	34.3	47.3	65.9
4B	0.2	4.6	19.0	38.8

Table 7: False “yes” on non-edges (% of finished answers), by number of shared neighbors in the true graph; all seven conditions pooled (including no primer and filler) at $p \leq .50$. Bin sizes (1.7B / 4B): 524/524, 350/350, 448/447, 728/727. Pooling across densities is descriptive.

Dense graphs (§4.3). Clustering raises dense plain 4B’s node-degree accuracy although its printed values barely vary there: at $p = .65$,

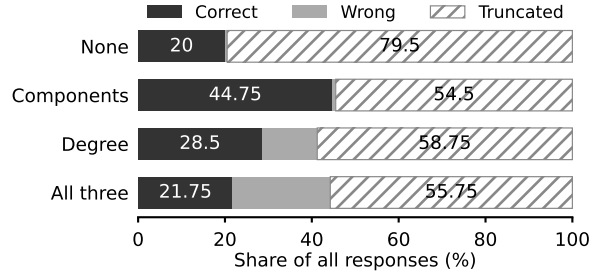


Figure 6: Thinking 4B on edge count: correct, wrong and truncated shares of *all* responses.

a graph’s clustering values have SD 0.021, and rounded RWSE takes 2.08 distinct value pairs per graph against 12.04 distinct degrees. Neither added length nor a change of strategy explains the gain: length-matched filler does not help (+2.0, $q = .59$), and neighbor enumeration is about as common with clustering as without (69% and 74%). The gain may therefore come from how the added numeric sentences change the prompt, which this design cannot isolate.

Against the thinking mode (§4.4). The one equivalence in Table 8 comes from the component sentence: on neighbor sets, the 90% interval of plain 4B’s gap to thinking 4B without a primer is $[-2.0, +2.5]$, within the prespecified ± 3 -point margin. For plain 1.7B’s neighbor sets under degree, the gap is also nonsignificant ($q = .24$), but its interval reaches +6.2, so equivalence is not established. To avoid selecting a primer on the graphs it is scored on, each row’s best primer is chosen on even-index graphs and scored on odd-index graphs, and vice versa; both halves chose the same primer.

Task	Model	No primer gap	Best primer for plain	Gap with best primer	Gap with Cl./RWSE
Degree	1.7B	+15.8	all	+5.5	+11.8
	4B	-2.2	components	-2.5	-1.8
Neighbors	1.7B	+6.2	degree	+2.8	+5.0
	4B	+9.5	components	+0.2 ^e	+8.5
Edge count [†]	1.7B	+14.5	degree	+11.5	+15.0
	4B	+18.0	degree	-9.2	+19.2
Edge exists	1.7B	+28.5	all	+17.0	+23.5
	4B	+9.2	clustering	+5.2	+5.2

Table 8: Gap in correct share (points): the same model thinking without a primer minus plain under no primer, the plain arm’s best primer selected on held-out graphs (see text), or the better of clustering and RWSE. Negative values put plain ahead. Bold: paired BH $q < .05$; [†]: at least 15% truncation on one side (Table 5), so no significance marks. Edge-existence gaps are in raw accuracy; balanced accuracy is in Table 6. ^e: the 90% interval lies within the prespecified ± 3 -point equivalence margin.

Thinking mode (§4.5). Under the degree primer, thinking 1.7B still lists the neighbors in 90% of responses at $p \geq .35$, and thinking 4B ends on the stated value in 93% of finished conflict responses, more often than thinking 1.7B. Because thinking responses are long, truncation drives many of the changes in these arms: of the answers a condition changes, either response hit the token budget in 44.4–64.5% for thinking 1.7B and 83.3–95.2% for thinking 4B (Figure 6 shows this for thinking 4B’s edge count).

Graph support (§4.6). Table 9 gives the values behind that section: no clustering or RWSE change in joint correctness is significant ($q \geq .16$). The table also reports valid cycle support.

Primer	Joint correct		Valid cycle support	
	1.7P	4P	1.7P	4P
none	54.00	85.75	34.2	58.5
degree	62.50	76.50	50.2	76.5
clustering	58.00	86.25	40.2	54.2
RWSE	58.25	83.75	32.0	33.2
all	61.75	77.50	38.8	45.2
components	54.50	95.00	29.8	63.7
filler	52.00	72.50	31.2	45.8

Table 9: Two graph-support checks in the plain arms (% of all prompts). *Joint correct*: both the degree and the exact neighbor set are right for the same queried node. *Valid cycle support*: a correct, finished cycle “yes” backed by a valid argument, a real cycle or the edge-count argument. Bold: BH $q < .05$ against no primer. Response categories are in Table 10.

Table 10 breaks every correct, finished cycle-check answer down by the argument it cites, and many rest on invented cycles. For example, on one $p = .20$ graph plain 1.7B without a primer answers with $0 \rightarrow 4 \rightarrow 1 \rightarrow 5 \rightarrow 14 \rightarrow 13 \rightarrow 15 \rightarrow 16 \rightarrow 18 \rightarrow \dots \rightarrow 38 \rightarrow 39 \rightarrow 0$, which climbs the ids almost in order; steps such as $15 \rightarrow 16$, $21 \rightarrow 25$ and $38 \rightarrow 39$ are not edges. The degree primer raises valid support for plain 4B as well as plain 1.7B (§4.6), while RWSE, all three statistics and filler reduce plain 4B’s. Thinking arms cite real cycles far more often (79.9% and 96.5% of correct answers without a primer), but RWSE turns 42.5% of thinking 4B’s correct answers into walks that are not cycles. Finally, truncation explains almost all of the degree primer’s cycle-check loss for thinking 1.7B, but less than half for plain 1.7B (Tables 3 and 5).

Arm	Primer	Correct	Cycle named			No cycle	
		“yes” (<i>n</i>)	Real	Invented	Not a cycle	Total	Edge count
1.7P	none	363	36.9	43.5	16.0	3.6	0.8
	degree	293	13.0	20.5	1.4	65.2	55.6
	clustering	397	39.5	48.1	7.8	4.5	1.0
	RWSE	362	33.7	40.3	19.9	6.1	1.7
	all	391	38.4	50.4	7.9	3.3	1.3
	components	371	27.8	40.4	25.3	6.5	4.3
	filler	375	33.3	55.5	9.6	1.6	0.0
1.7T	none	388	79.9	17.8	1.0	1.3	0.5
	degree	301	61.5	29.6	0.0	9.0	9.0
	clustering	386	80.6	19.2	0.0	0.3	0.3
	RWSE	397	68.0	20.7	10.6	0.8	0.5
	all	385	73.0	21.8	3.1	2.1	1.8
	components	358	74.9	22.6	0.0	2.5	2.0
	filler	400	77.2	19.8	2.5	0.5	0.0
4P	none	396	34.1	23.0	6.8	36.1	25.0
	degree	387	13.4	3.9	2.1	80.6	65.6
	clustering	398	48.7	31.2	1.0	19.1	5.8
	RWSE	395	33.2	20.8	38.7	7.3	0.5
	all	395	45.1	33.4	6.6	14.9	0.8
	components	388	39.2	18.6	3.4	38.9	26.5
	filler	394	44.2	28.2	5.3	22.3	2.3
4T	none	397	96.5	3.3	0.0	0.3	0.0
	degree	375	90.9	5.1	0.0	4.0	2.7
	clustering	397	95.5	4.5	0.0	0.0	0.0
	RWSE	398	55.0	2.5	42.5	0.0	0.0
	all	395	92.7	6.3	0.0	1.0	1.0
	components	384	92.2	4.4	0.0	3.4	2.9
	filler	399	94.2	5.0	0.5	0.3	0.0

Table 10: What the correct cycle-check answers rest on (main sweep; every correct answer is “yes”). *Correct (n)*: count of correct, finished answers; the other columns are % of those *n*, and Real, Invented, Not a cycle and Total sum to 100. *Real*: every step of the cycle is an edge; *Invented*: a step is not an edge; *Not a cycle*: a length-2 or retraced walk. *Edge count*: the answers under Total that argue from the number of edges, identified by their wording. The valid arguments are Real and Edge count, which is valid because 40 nodes with at least 40 edges must contain a cycle. Bold (Real, Invented only): differs from the same model without a primer (paired, BH $q < .05$).

Node count. This task cannot show graph reading, because every graph has 40 nodes, all named in the encoding’s first line. Plain 1.7B often mistakes the last identifier, 39, for the count, and most primers reduce that error: clustering nearly eliminates it (68.5% to 1.2% of finished answers), and even filler, which states nothing, raises accuracy by 39.8 points. The large effect in Table 3 is therefore not specific to clustering and does not show graph reading.

C Pilot and the Clustering Screen

Pilot. On the published GraphQA graphs (5–19 nodes), Gemma 4 E4B/12B (Gemma Team, 2026) and Qwen3-8B/14B, in both modes, were near ceiling without a primer on most tasks. Edge count remained difficult. These pilot results motivated the 40-node graphs and smaller models; they are not pooled with the main sweep.

The clustering effect. Because clustering’s gain on plain 1.7B’s node degree emerged from a 40-cell screen, Table 11 retests it. The unscreened, fresh-seed, fixed-mean-degree and Qwen3-8B rows use graphs the screen did not see; the $p \leq .50$, against-filler and $p \geq .65$ rows use the full 400-graph-per-density runs, which include the screened graphs. The screened-graph row is descriptive only, because it reuses graphs from the initial search. The gain holds up on unscreened graphs, a fresh seed and the fixed-mean-degree grid, where it is largest in the densest cell (20 nodes, $p = .842$: +16.5) but reverses in the other 20-node cell ($p = .421$). However, it is not detected on 40-node graphs at $p \geq .65$, nor in Qwen3-8B, which is near ceiling in most cells (Table 11). Thinking 1.7B shows a similar split in the follow-ups: clustering gains it 2.5 points at $p \leq .50$ ($q = .043$) but loses 5.4 at $p \geq .65$ ($q = .0015$), where filler loses 11.4.

Test	N	Gain	Statistic	Correction
Screened graphs	700	+1.29	$p = .47$	none
Unscreened graphs	2,100	+2.1	$p = .031$	none
$p \leq .50$	1,600	+3.8	$q = .0035$	BH
Fresh seed	1,600	+4.2	$p = .00055$	Bonf. = .022
Against filler	1,600	+3.2	$q = .014$	BH
Fixed mean degree	3,200	+6.2	$q < .001$	one test
$p \geq .65$	1,200	−0.7	$q = .57$	BH
Qwen3-8B	3,200	−1.0	$q = .05$	one test

Table 11: Node-degree accuracy change (points) for plain Qwen3-1.7B, except the Qwen3-8B row. N is paired prompts. Comparisons use no primer unless labeled against filler. *Screened*: the 700 graphs the discovery screen used; *unscreened*: the other 2,100; *fresh seed*: newly generated graphs; *fixed mean degree*: the 20–160-node grid. p is unadjusted; q is BH-adjusted (equal to p in a one-test family); the fresh-seed Bonferroni value corrects across the 40 screened cells.

How clustering helps is unclear. No sampled response explicitly invokes clustering, and the gain does not vary detectably with the queried node’s coefficient ($p = .50$), though it grows with the node’s degree ($p = .012$). The clearest pattern is list position: the gain is +5.01, +0.75 and −0.11 points for queried nodes in the first, middle and

last list thirds, and, adjusting for degree, clustering and density, it shrinks by 1.74 points per ten later positions ($p = .018$), while filler and component controls show no corresponding gradient. These patterns motivate a position-based explanation but do not prove one: clustering resolves some prior copying errors while introducing others, and components and filler also reduce missed neighbors, so no clustering-specific mechanism is isolated. Shuffling node order would test the position hypothesis directly.